Reda El Ferrasi

Junior Machine Learning Engineer

+212 607420430 | redaelferrasi@gmail.com | linkedin.com/in/reda-elferrasi/ | github.com/redaelfer

PROFILE

State-qualified engineer in Data Science and IoT, with projects classifying images and security logs. Experience in data preparation, transfer learning and model integration through Flask APIs.

TECHNICAL SKILLS

Programming: Python.
ML: TensorFlow/Keras, scikit-learn, **XGBoost**.
Data: Pandas, NumPy.
Tools: **Flask**, Docker, Git.
Languages: Arabic (native), English (B2), French (B2).

PROFESSIONAL EXPERIENCE

AI Engineer | Matious Digital – Final-year project (PFE) Jan. 2026 – Sep. 2026

- Developed and tested a platform with **20 to 40 concurrent AI agents**.
- Integrated **OpenClaw and Hermes Agent** through WebSocket and REST APIs.
- Automated prompt generation, model selection, routing and error recovery.
- Deployed containerized agents on Hetzner Cloud.

Data Engineering Intern | ONCF Jun. 2025 – Aug. 2025

- Automated SNMP/Syslog ETL: reduced latency by **23 %** during testing.
- Centralized data and created dashboards with **more than 10 KPIs**.
- Reduced incident detection time by **35 %** during NOC testing.

Full Stack Developer Intern | S4U Jul. 2024 – Aug. 2024

- Developed a React ticketing SaaS: reduced processing time by **30 %** in tests.
- Added Jest tests with **72 % coverage** of key features.

PROJECTS

RetinaScan | Medical AI | github.com/redaelfer/RetinaScan

- Adapted pretrained **MobileNetV2** for retinal image classification, with a frozen base.
- Achieved **approximately 74 % validation accuracy** in **10 epochs**, with image augmentation and **dropout of 0.5**.
- Saved the **TensorFlow** model and integrated its predictions into the application through a **Flask API**.

APT Attack Detection | AI and Cybersecurity | github.com/redaelfer/NosqlAPT

- Prepared **DAPT 2020** (Kaggle) logs: missing values, encoding and normalization.
- Combined **autoencoder** representations and **XGBoost**, with early stopping.
- Recorded an **internal validation AUC of 0.984**; tracked precision, recall and F1.

EDUCATION

ENSIAS – State Engineering Degree – Data Science & IoT 2023 – 2026

CPGE – TSI preparatory classes 2021 – 2023

CERTIFICATIONS & INTERESTS

CS50x | Harvard University

AWS Knowledge: Cloud Essentials – Training badge | AWS

Interests: ECU remapping, car/motorcycle mechanics and repair, game modding, history/mythology.